

Unraveling the Story of the Apollo Lunar Thin Section 76539-10-r1

Ping Wang

September 2025

1 Introduction

This lunar rock 76539 is a small aphanitic basalt found in the rake sample collected at station 6, Apollo 17. The word *aphanitic* means fine-grained. Seen from the following image of sample 76539, this lunar rock is so fine-grained that individual mineral crystals cannot be seen with the naked eye. An aphanitic texture forms when magma cools rapidly, preventing crystals from growing large.



Figure 1: Apollo lunar sample 76539 from Apollo 17. Credit: Virtual Microscope

Paired rotational series of PPL and XPL images of sample 76539-10-r1 at 5° intervals was acquired from Virtual Microscope [1] and subsequently aligned to ensure a consistent field of view.



Figure 2: Sample 76539-10-r1 in PPL (top) and XPL (bottom). Field of view 2.0 mm

2 Petrographic Observations

2.1 Skeletal grains

Several distinct skeletal grains are present:

Elongate chain /swallowtail crystal. The most conspicuous grain is a long, narrow crystal crossing the upper-central part of the field on a diagonal. It is not a solid blade: it shows a segmented, internally hollow structure and a forked (swallowtail) termination.

Hopper (hollow-box) crystals. At least one well-defined hollow, box-shaped cross-section sits on the right side of the field, and additional hollow frames appear in the lower-left cluster. These are hopper sections: in PPL, they read as bright rectangular outlines enclosing dark centers; in XPL, the crystalline walls take on moderate-to-high interference colors (the turquoise, gold, tan, lavender).

Open lattice /grid frameworks. The lower-left quadrant contains several crystals built from parallel and cross-connected bars, producing a grid- or ladder-like appearance. This is the skeletal end of the morphology spectrum, intermediate between hopper and fully dendritic growth.

Interpretation:

Elongation with hollow segments and bifurcating tips is diagnostic of growth at high undercooling, where extension along a fast-growth direction outpaces infilling of the crystal faces. Hopper sections infer that the crystal edges and corners are developed while the interior remains an open cavity, subsequently occupied by glass.

2.2 Groundmass

In a lunar thin section, groundmass is the fine-grained assemblage of materials and /or glass that surrounds the larger crystals; whereas phenocrysts are the large, early-formed crystals surrounded by a fine-grained groundmass.

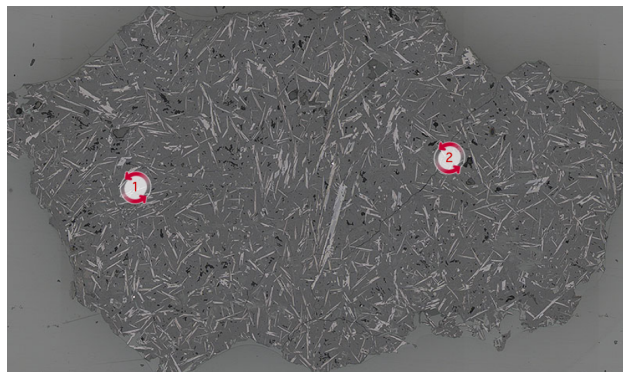


Figure 3: Sample 76539 under reflected light

Seen from the relected light image of lunar rock 76539,

3 Segmentation of Opaques

You may want to give it a try to segment opaque phases in the PPL and XPL images, allowing the texture of the sample to be visualized more clearly: <https://huggingface.co/spaces/pingwang2026/segmentation-of-opaques-in-petrographic-images>

The Moon may seem still and cool, but it was once thermally active, and alive with molten rock flowing across its surface and shaping its geology.

References

- [1] "76539 Vitrophyric basalt," *The Virtual Microscope*, The Open University. [Online]. Available: <https://www.virtualmicroscope.org/content/76539-10-vitrophyric-basalt>
- [2] W. D. Nesse, *Introduction to Optical Mineralogy*, 4th ed. New York, NY, USA: Oxford University Press, 2014.
- [3] W. A. Deer, R. A. Howie, and J. Zussman, *An Introduction to the Rock-Forming Minerals*, 3rd ed. London, U.K.: Mineralogical Society, 2013.
- [4] C. Meyer, *Lunar Sample Compendium*. Houston, TX, USA: NASA Johnson Space Center, 2011. [Online]. Available: <https://curator.jsc.nasa.gov/lunar/lsc/>
- [5] J. M. Rhodes, N. J. Hubbard, H. Wiesmann, K. V. Rodgers, J. C. Brannon, and B. M. Bansal, "Chemistry, classification, and petrogenesis of Apollo 17 mare basalts," in *Proceedings of the 7th Lunar Science Conference*, Houston, TX, USA, 1976, pp. 1467–1489.
- [6] T. M. Usselman, G. E. Lofgren, C. H. Donaldson, and R. J. Williams, "Experimentally reproduced textures and mineral chemistries of high-titanium mare basalts," in *Proceedings of the 6th Lunar Science Conference*, Houston, TX, USA, 1975, pp. 997–1020.
- [7] C. H. Donaldson, "An experimental investigation of olivine morphology," *Contributions to Mineralogy and Petrology*, vol. 57, no. 2, pp. 187–213, 1976.